

Exercise on MESI protocol and consistency

A processor has two cores (P0 and P1), each one with a L1 direct-mapped private cache with 8 lines and a block size of 16 B. The coherence protocol is MESI. Show the evolution of the MESI states and of the caches.

```
.data 0x0120
X: .word 10

.text
# P0
lw $t0, 0x0120($zero)
li $t1, 20
sw $t1, 0x0120($zero)
lw $t2, 0x0120($zero)

# P1
lw $s0, 0x0120($zero)
lw $s1, 0x0120($zero)
```

Assume that the order of execution is interleaved, in terms of memory operations, between P0 and P1.

Partial Solution (only MESI states evolution is shown)

4 bits offset, 3 bits tag and 9 bits tag.

```
P0: lw $t0, 0x0120($zero) -> E
P1: lw $s0, 0x0120($zero) -> S
P0: sw $t1, 0x0120($zero) -> M e I
P1: lw $s1, 0x0120($zero) -> S
P0: lw $t2, 0x0120($zero) -> local hit
```